# MART-NILM: A Compact Multi-Appliance Relational Temporal Network for Joint Power–State Estimation in Non-Intrusive Load Monitoring

Raymond Tie Siou Hin, Tan Wen Shan, Ding Ze Yang, Mirza Rayana Sanzana, and Haval Kamil

***Abstract*—Appliance-level load visibility supports residential demand-side management, but aggregate meters do not directly reveal individual consumption or operating states. Non-intrusive load monitoring (NILM) estimates these quantities without dedicated appliance sensors. Multi-appliance disaggregation shares computation across loads, yet sparse activations under limited background conditions and unequal power–state loss scales complicate joint learning. This paper proposes the Multi-Appliance Relational Temporal Network for NILM (MART-NILM), a compact joint estimator. A shared temporal convolutional encoder, appliance-specific decoders, and relational attention jointly estimate five appliance powers and states. Focal-event background-stratified mixing exposes active appliance traces to varied residual loads, while bounded appliance-wise balancing adjusts each classification objective relative to its corresponding regression loss. Both strategies operate only during training. Experiments on two public residential datasets use disjoint training, validation, and test houses and preserve recorded aggregates. Across three seeds, MART-NILM achieves appliance-average mean absolute errors of 7.3 and 10.9 watts and macro-averaged F1 scores of 0.88 and 0.72 on the respective test houses, outperforming three reproduced baselines on both measures. With 1.373 million parameters and lower training time under the evaluated configurations, the method offers a favorable estimation–cost trade-off for residential load monitoring.**

***Index Terms*—non-intrusive load monitoring, energy disaggregation, multi-task learning, power and state prediction, household appliances**

## I. Introduction

Residential demand-side management requires appliance-level observability of both demand magnitude and operating activity. Power estimates quantify appliance demand, whereas state estimates expose activation timing; aggregate metering reveals neither directly. Appliance-level feedback can support energy conservation [1], but dedicated submetering requires installing and maintaining additional sensors. Non-intrusive load monitoring (NILM) estimates appliance consumption from aggregate measurements and offers a less instrument-intensive basis for residential load analysis [2]. Joint estimation is important because low average power error can conceal missed operating periods, particularly for sparsely active appliances.

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Classical NILM identifies appliance signatures and infers appliance states with finite-state or factorial hidden Markov models [3], [4]. Deep approaches instead learn representations directly from aggregate windows [5], reducing the need to specify signatures and state-transition models manually. Regardless of the estimator, a recorded aggregate contains demand outside the monitored target set. For $N$ target appliances, the measured power is

$$x_t = \sum_{i=1}^{N} P_{t,i} + b_t + \epsilon_t, \tag{1}$$

where $P_{t,i}$ is target power, $b_t$ is unmonitored household demand, and $\epsilon_t$ represents measurement and timing discrepancies. Because residual demand can exceed a low-power target and varies across households, it creates a deployment-domain shift beyond changes in appliance signatures. House-held-out evaluation on recorded aggregates is therefore adopted here to assess cross-house generalization. Results obtained under different sensing and partitioning protocols are not directly interchangeable [6].

Within this setting, two common deep NILM modelling paradigms are single-appliance and multi-appliance estimation. Single-appliance networks predict one target per model and have advanced disaggregation accuracy [7], [8], but repeating a model for every load duplicates aggregate feature extraction and storage, which is undesirable for smart-meter or home-gateway deployment. Recent deep NILM work has therefore increasingly studied multi-appliance estimation, in which one model jointly estimates a fixed set of $N$ target appliances from the same aggregate window [9]–[12]. NILM research commonly considers two complementary tasks: power estimation and ON–OFF state detection. For appliance $i$ at time $t$, these tasks estimate a power trace $\hat{y}_{t,i}$ and an ON-state probability $\hat{s}_{t,i}$, respectively. Power estimation is a regression task, whereas state detection is multi-label classification in the multi-appliance setting because several appliances may be ON at the same time. For an aggregate window $\mathbf{x}$ of length $T$, the joint estimator is

$$\left(\hat{\mathbf{Y}}, \hat{\mathbf{S}}\right) = f_{\boldsymbol{\theta}}(\mathbf{x}), \qquad \hat{\mathbf{Y}} \in \mathbb{R}^{T\times N}, \quad \hat{\mathbf{S}} \in [0,1]^{T\times N}, \tag{2}$$

where $\boldsymbol{\theta}$ denotes the model parameters, and $\hat{\mathbf{Y}}$ and $\hat{\mathbf{S}}$ collect the estimated power traces and ON-state probabilities, respectively.

Joint multi-appliance and multi-label estimation is more challenging than running $N$ independent predictors. Concur-

rent activations provide useful context, but superposed transitions and residual demand can mask weak targets [13]. A joint architecture must therefore separate appliance-specific evidence while retaining relational context and coherent power–state outputs.

Training raises two further difficulties. First, sparse events in a finite record occur under only a subset of residual-background conditions. A model can then use residual-background correlations as a shortcut instead of the appliance signature; this can fail when those correlations change across households [13]. Second, appliance power ranges and on-state prevalences differ substantially. Existing joint objectives use a shared regression–classification balance, either as a fixed coefficient or a global loss-ratio weight [12], [14]. Such a shared balance cannot represent appliance-specific scales; when derived from joint losses, a large power error for one appliance also changes the relative state weight of the others. These observations motivate control over both residual-background conditions and the power–state balance of each appliance.

To address these issues, we propose the Multi-Appliance Relational Temporal Network for NILM (MART-NILM), a relational temporal model for joint power and ON–OFF state estimation under residual-background variation.

The main contributions are threefold:

1) A compact multi-appliance architecture, MART-NILM, is designed for joint power and state estimation. It extracts a shared temporal representation once and combines appliance-specific decoding with cross-appliance relational context to produce five paired power–state outputs in a single forward pass.
2) Two training-only strategies address sparse events, residual-background variation, and appliance-wise power–state imbalance. Focal-event background-stratified mixing exposes activations to a controlled range of recorded residual-background conditions, while bounded appliance-wise balancing independently aligns the power and state objectives of each appliance. Neither strategy adds inference-time computation.
3) A house-held-out evaluation uses recorded aggregates from two public residential datasets. Three-seed baseline comparisons, architecture and training ablations, and background-conditioned analyses quantify power estimation, state detection, computational cost, and remaining failure modes.

## II. Related Work

Multi-appliance and multi-label NILM estimates power and predicts an ON–OFF label for each appliance in a fixed set from a single aggregate measurement. This setting has a nested multi-task structure: the inner level couples power regression and state classification for each appliance, whereas the outer level determines how representations are shared across appliances that may be active concurrently. Existing NILM studies recognize these two levels through formulations described as "multi-task-multi-appliance" [9] and "multi-appliance-task" [12], [15].

Generic multi-task learning (MTL) commonly treats tasks as a peer set and learns which parameters or intermediate representations to share across them [16]. Joint multi-appliance and multi-label NILM must instead preserve coupled power–state objectives within each appliance while separating and relating appliances that may be active concurrently.

Relevant NILM studies, including multi-appliance and multi-label formulations, use convolutional neural networks (CNNs) [9], [15], recurrent neural networks (RNNs) [17], temporal convolutional networks (TCNs), including attention-enhanced variants [10], [18], and Transformers [12]. Backbone choice controls temporal feature extraction; the remaining architectural question is how those features are organized for joint prediction. In joint multi-appliance and multi-label estimation, this organization entails two linked design decisions: *appliance-head design*, which governs within-appliance power–state output coupling, and *cross-appliance body design*, which governs where appliance representations separate, remain shared, or exchange information before reaching the appliance heads. Figures 1 and 2 summarize these two levels, respectively.

At the appliance-head level, early joint NILM designs use separate state and power branches, with the predicted state concatenated with power-branch features before the final regression layers. The CNN and RNN models of Murray *et al.* [17], illustrated in Fig. 1(a), use this state-informed concatenation. It allows the regressor to use state information while retaining direct power features, but does not explicitly enforce consistency between OFF predictions and power estimates.

An explicit connection is introduced by the subtask-gated network (SGN) [19], which multiplies the regression output by a separately supervised on-state probability, as shown in Fig. 1(b). This gate suppresses power during predicted inactivity, although a missed activation also attenuates an otherwise valid power estimate. The basic formulation suppresses standby consumption, motivating a standby-power variant in the original study [19].

To accommodate different event durations and temporal contexts, the scale- and context-aware network (SCANet) [20] extends SGN with parallel dilated branches and self-attention. As summarized in Fig. 1(c), classification features gate the corresponding regression features at each scale before feature fusion and contextual processing, while the final output retains state gating. The design supports multi-scale feature selection but adds branching and attention cost and remains sensitive to state-derived gates. These appliance-head mechanisms can be combined with different ways of sharing information across appliances.

The resulting design question is where to share information across appliances. Multi-task architectures commonly share an encoder and retain task-specific outputs, or continue exchanging information during decoding [16]. Multi-Appliance-Task NILM (MATNilm) explicitly branches by appliance and then by regression and classification task [15]; Fig. 2(a) isolates this hierarchical branch pattern. Shared encoding reduces repeated feature extraction, while private branches retain capacity for different appliance signatures.

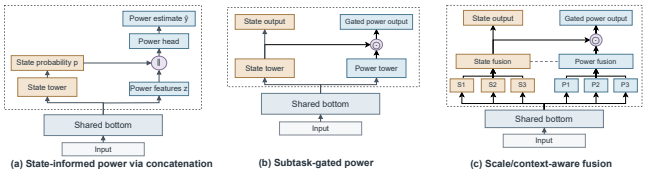



Fig. 1. Within-appliance power–state heads: (a) concatenation [17], (b) output gating [19], and (c) multi-scale fusion [20].

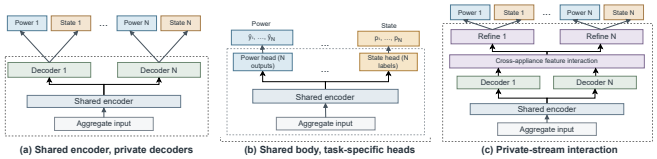



Fig. 2. Multi-appliance body designs: (a) shared encoder/private decoders [15], (b) shared body/task heads [9], and (c) private-stream interaction.

A fully shared body with task-specific output heads is another joint-estimation design. As shown in Fig. 2(b), it retains one representation for all appliances, then uses separate heads to predict the vector of appliance powers and the vector of independent appliance-state labels. UNet-NILM is one example of this pattern [9].

Feature interaction, illustrated in Fig. 2(c), provides a complementary mechanism. In dense prediction, attention-guided message passing transfers information from intermediate task predictions to final tasks [21]. In NILM, MATNilm already models dependencies across appliances and time [15]. This principle motivates interaction between appliance-specific feature streams.

Architecture alone does not determine joint learning. Gradient normalization balances task updates directly [22], whereas SAMNet uses a power–state loss ratio [14]. Background recomposition addresses a different issue: exposure to residual demand beyond the target set [13]. Together, these studies motivate evaluating architectural sharing, appliance-wise power–state balance, and residual-background coverage.

## III. Methodology

### A. Problem Formulation

Following (1), the model estimates the paired power and ON–OFF state outputs of $N$ target appliances from an aggregate window $\mathbf{x}$ of length $T$:

$$f_{\boldsymbol{\theta},t}(\mathbf{x}) = \underbrace{\big(\hat{y}_{t,1}, \hat{s}_{t,1}, \ldots, \hat{y}_{t,N}, \hat{s}_{t,N}\big)}_{N \text{ appliance power–state pairs}}, \quad t = 1, \ldots, T. \tag{3}$$

Here $\hat{y}_{t,i}$ estimates appliance power on the standardized scale, and $\hat{s}_{t,i} \in [0, 1]$ is its ON-state probability. Power estimation is a regression task, whereas state detection is multi-label classification because several appliances can be ON simultaneously. Binary ON–OFF labels summarize activity without resolving internal operating modes.

### B. Model Architecture

The implementation uses instance–batch normalization (IBN), batch normalization (BN), the rectified linear unit (ReLU) activation, and fully connected (FC) output layers.

Figure 3 presents MART-NILM, combining the private-stream interaction body in Fig. 2(c) with the state-gated heads in Fig. 1(b). The components are described below in processing order, using the same names as the architecture ablations in Table V.

*Multi-scale CNN:* Parallel convolutional branches [23] extract local features at complementary time scales from the standardized aggregate $\bar{\mathbf{x}}$:

$$\mathbf{Z} = \mathcal{F}([\mathcal{B}_3(\bar{\mathbf{x}}); \mathcal{B}_5(\bar{\mathbf{x}}); \mathcal{B}_9(\bar{\mathbf{x}})]) + \mathcal{S}(\bar{\mathbf{x}}). \tag{4}$$

Here $\mathcal{B}_k$ is a kernel-$k$ convolution followed by IBN [24] and ReLU; $[\cdot;\cdot]$ denotes channel concatenation. The $1 \times 1$ fusion $\mathcal{F}$ combines the branches, while the projected shortcut $\mathcal{S}$ retains a direct input path. IBN applies instance normalization (IN) and BN to separate channel halves.

*Feature encoder:* Two convolution–IBN–ReLU stages combine the local features into a shared representation $\mathbf{H}^{(0)} \in \mathbb{R}^{128 \times T}$. Feature extraction is performed once for all appliances.

*Temporal encoder:* Residual dilated TCN blocks [25] extend temporal context:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathcal{T}_{2^\ell}(\mathbf{H}^{(\ell)}), \qquad \ell \in \{0, 1, 2, 3, 4\}. \tag{5}$$

Here $\mathcal{T}_d$ comprises a dilation-$d$ convolution, BN, ReLU, and dropout. Increasing dilation expands context, while residual connections and symmetric padding retain the incoming features and temporal alignment. The output is $\mathbf{H} = \mathbf{H}^{(5)}$.

*Appliance decoders:* Each decoder applies a channel gate, inspired by channel recalibration [26], and a local residual transformation to obtain an appliance-specific representation $\mathbf{F}_i$. These private streams adapt the shared temporal features to individual load characteristics.

*Relation attention:* The relation module draws on the cross-task information-sharing principle of PAD-Net [21] to exchange context among appliance-specific feature streams. Scaled dot-product attention [27] is evaluated across appliances at each time $t$:

$$\mathbf{A}_t = \mathrm{softmax}\left(\frac{\mathbf{Q}_t \mathbf{K}_t^{\mathsf{T}}}{\sqrt{d}}\right), \qquad \mathbf{M}_t = \mathbf{A}_t \mathbf{V}_t. \tag{6}$$

Here $\mathbf{Q}_t$, $\mathbf{K}_t$, and $\mathbf{V}_t$ are learned projections of the stacked appliance representations, and $d$ is their projection width. The attended context is projected back to the feature dimension and added to each private stream through a gated residual connection.

*Joint power and state prediction:* For each appliance, two time-shared FC heads predict raw standardized power $\hat{y}_i^{\mathrm{raw}}(t)$ and a state logit $a_i(t)$. Following the state-gating principle of SGN [19], the on-state probability $\hat{s}_i(t) = \mathrm{sigmoid}(a_i(t))$ softly gates the power prediction:

$$\hat{y}_i(t) = \hat{s}_i(t)\hat{y}_i^{\mathrm{raw}}(t) + [1 - \hat{s}_i(t)]y_i^{(0)}, \tag{7}$$

where $y_i^{(0)} = -\mu_i/\sigma_i$ is the standardized zero-power anchor. Hence, a low on-state probability smoothly moves the estimate

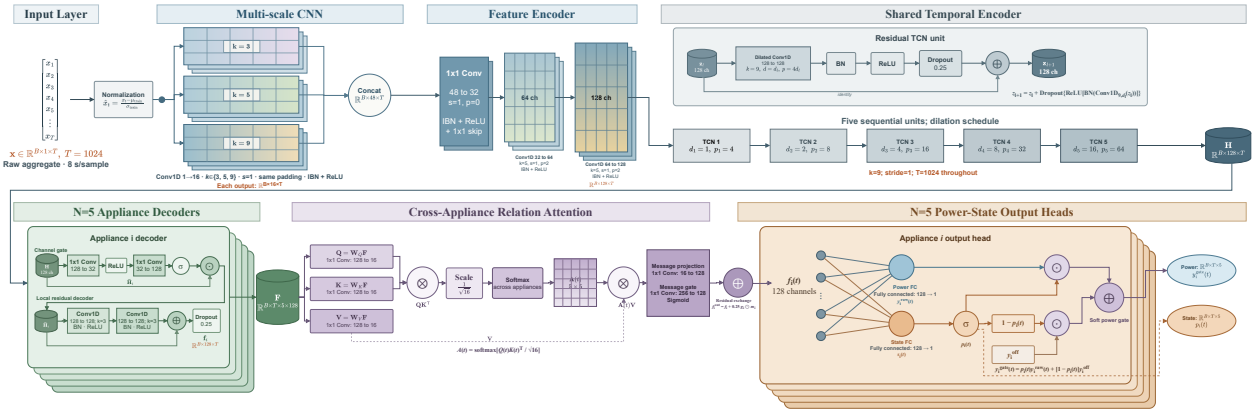



Fig. 3. Model architecture of MART-NILM.

towards zero power. Predictions are converted back to watts using the appliance training mean $\mu_i$ and standard deviation $\sigma_i$, and negative values are clipped to zero. The gate couples the state and power predictions while retaining a separate waveform regressor.

### C. Loss Function

The model is trained jointly over all appliances, as in multi-task NILM [9], [12], [15]. Let $\mathcal{L}_{\text{power},i}$ and $\mathcal{L}_{\text{state},i}$ denote the power and state losses for appliance $i$, respectively. The joint objective is

$$\mathcal{L} = \sum_{i=1}^{N} \left(\mathcal{L}_{\text{power},i} + \lambda r_i \mathcal{L}_{\text{state},i}\right), \qquad \lambda = 0.8. \tag{8}$$

Here $r_i$ is an appliance-wise balancing factor defined below.

*Power loss:* The power objective combines waveform fidelity and switching fidelity:

$$\mathcal{L}_{\text{power},i} = \underbrace{\mathcal{L}_{\text{wave},i}}_{\text{power-level fidelity}} + \underbrace{0.15\,\mathcal{L}_{\text{trans},i}}_{\text{transition fidelity}}. \tag{9}$$

The waveform and transition terms both use mean squared error (MSE), applied to power values and temporal differences, respectively. The expressions below are written for one window; in implementation, both terms are averaged over the batch and time dimensions. Let $y_i(t)$ and $\hat{y}_i(t)$ denote standardized target and predicted power, and let $z_i(t) \in \{0,1\}$ denote the on–off label. For a temporal index set $\mathcal{S}$, define

$$\text{MSE}_{\mathcal{S}}(u,v) = \frac{1}{\max(|\mathcal{S}|,1)} \sum_{t\in\mathcal{S}} \big(u(t)-v(t)\big)^2. \tag{10}$$

The waveform term evaluates power errors over all samples $\mathcal{T}_0 = \{1,\dots,T\}$, on-state samples $\mathcal{T}_i^{\text{on}} = \{t \in \mathcal{T}_0 : z_i(t) = 1\}$, and off-state samples $\mathcal{T}_i^{\text{off}} = \{t \in \mathcal{T}_0 : z_i(t) = 0\}$:

$$\begin{aligned}\mathcal{L}_{\text{wave},i} = {} & \text{MSE}_{\mathcal{T}_0}(y_i,\hat{y}_i) + \text{MSE}_{\mathcal{T}_i^{\text{on}}}(y_i,\hat{y}_i) \\ & + 0.5\ \text{MSE}_{\mathcal{T}_i^{\text{off}}}(y_i,\hat{y}_i).\end{aligned} \tag{11}$$

The first term measures global waveform fidelity; the on-state term prevents sparse activations from being dominated by off samples, and the lower-weight off-state term constrains inactive predictions.

To additionally constrain changes in power, let $\Delta y_i(t) = y_i(t) - y_i(t-1)$ and define $\Delta\hat{y}_i(t)$ analogously. The transition term compares these differences whenever the appliance is on at either adjacent sample, over $\mathcal{T}_i^{\Delta} = \{t \in \{2,\dots,T\} : z_i(t) = 1 \text{ or } z_i(t-1) = 1\}$:

$$\mathcal{L}_{\text{trans},i} = \text{MSE}_{\mathcal{T}_i^{\Delta}}(\Delta y_i, \Delta\hat{y}_i). \tag{12}$$

*State loss:* The state objective combines weighted binary cross-entropy (BCE) with explicit false-positive control:

$$\mathcal{L}_{\text{state},i} = \underbrace{\mathcal{L}_{\text{BCE},i}}_{\text{class-balanced state detection}} + \underbrace{\mathcal{L}_{\text{fp},i}}_{\text{false-positive control}}. \tag{13}$$

For a mini-batch of $B$ windows, let $\hat{s}_{b,t,i} \in (0,1)$ be the predicted on-state probability and $z_{b,t,i} \in \{0,1\}$ the target label. The class-balanced BCE term uses $w_i = \min\{(1-\pi_i)/\pi_i, 12\}$, determined by the training on-state proportion $\pi_i$:

$$\begin{aligned}\mathcal{L}_{\text{BCE},i} = -\frac{1}{BT} \sum_{b,t} \big[ & w_i z_{b,t,i} \log \hat{s}_{b,t,i} \\ & + (1 - z_{b,t,i}) \log(1 - \hat{s}_{b,t,i}) \big].\end{aligned} \tag{14}$$

The false-positive term penalizes confident on-state predictions on off samples:

$$\mathcal{L}_{\text{fp},i} = \frac{\sum_{b,t}(1 - z_{b,t,i})\hat{s}_{b,t,i}^2}{\max\left(\sum_{b,t}(1 - z_{b,t,i}), 1\right)}. \tag{15}$$

*Appliance-wise task balance:* To account for the different numerical scales of regression and classification losses, a bounded loss-ratio weighting is used. It is motivated by loss-ratio balancing in SAMNet [14] and differs from gradient-norm balancing in GradNorm [22]. The global reference ratio $g$ bounds the appliance-specific ratio $r_i$:

$$\begin{aligned} g &= \frac{\sum_j \mathcal{L}_{\text{power},j}}{\sum_j \mathcal{L}_{\text{state},j}}, \\ r_i &= \min\left\{3g, \max\left\{\frac{g}{3}, \frac{\mathcal{L}_{\text{power},i}}{\mathcal{L}_{\text{state},i}}\right\}\right\}. \end{aligned} \tag{16}$$

At each mini-batch update, $g$ and $r_i$ are evaluated from the current losses and held fixed during back-propagation. The three-fold bounds limit each local ratio to $[g/3, 3g]$, preventing extreme weights from dominating optimization.

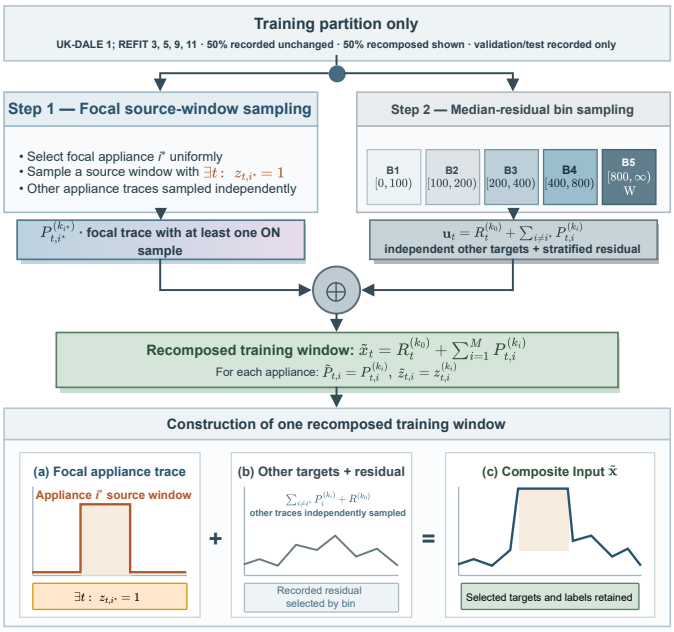



Fig. 4. Focal-event stratified mixing recomposes $\tilde{x}_t$ from an on-event focal trace, a power-stratified residual background, and independently sampled remaining appliance traces.

### D. Focal-Event Background-Stratified Mixing

Training uses recorded and recomposed aggregate windows with equal probability. Recorded windows preserve the observed joint demand. Recomposed windows extend synthetic aggregate construction in Neural NILM [5] by pairing appliance events with recorded residual backgrounds from other training windows [13], reducing reliance on house-specific event–background correlations. For a recorded sequence, the non-target residual is

$$R_t = \left[ x_t - \sum_{i=1}^{N} P_{t,i} \right]_+ . \tag{17}$$

For each recomposed window, the residual and appliance traces are drawn from independently selected training windows:

$$\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{N} P_{t,i}^{(k_i)}, \qquad \tilde{P}_{t,i} = P_{t,i}^{(k_i)}, \quad \tilde{z}_{t,i} = z_{t,i}^{(k_i)}, \tag{18}$$

where $k_0, k_1, \ldots, k_N$ denote independent window indices. One focal appliance $i^\star$ is selected uniformly, and its trace is drawn from a window containing at least one on sample. The residual is sampled from a uniformly selected non-empty bin of median residual power, using $[0, 100)$, $[100, 200)$, $[200, 400)$, $[400, 800)$, and $[800, \infty)$ W; all remaining appliance traces are sampled independently. Thus, rare events occur against varied recorded backgrounds while their selected power and state labels are retained. Recomposition is used only during training; validation and test use recorded aggregates and aligned labels.

## IV. Experimental Evaluation

### A. Dataset

Two public residential datasets are used: the UK Domestic Appliance-Level Electricity (UK-DALE) dataset [28] and the REFIT electrical-load dataset [29]. The target appliances are the kettle, fridge, dishwasher, washer, and microwave, and all signals are resampled at 8 s.

The experiment uses a whole-house split: UK-DALE house 1 and REFIT houses 3, 5, 9, and 11 for training; UK-DALE house 5 and REFIT house 2 for validation; and UK-DALE house 2 and REFIT house 20 for testing. Each house contributes one continuous 5-week block. The held-out blocks are centred within the previously established eight-week recording intervals and are selected without reference to appliance events or model performance. Normalization statistics are estimated from the training houses only.

### B. Data Preprocessing

After 8 s resampling, no additional temporal alignment is applied; residual timing discrepancies remain a measurement limitation. Rows missing any required aggregate, appliance-power, or state-label value are excluded, and windows do not cross timestamp discontinuities.

Let $n$ index windows and $t$ index samples within a window. The aggregate mean and standard deviation, $\mu_x$ and $\sigma_x$, and the corresponding appliance-$i$ statistics, $\mu_i$ and $\sigma_i$, are estimated from the training houses. The input and target powers are standardized as

$$\bar{x}_{n,t} = \frac{x_{n,t} - \mu_x}{\sigma_x}, \qquad y_{n,t,i} = \frac{P_{n,t,i} - \mu_i}{\sigma_i}. \tag{19}$$

State labels are precomputed before windowing. For appliances marked in Table I, isolated spikes are first removed on a label-only copy of the resampled power trace. Each sample is then labeled on when this trace reaches its appliance-specific threshold $\tau_i^P$. OFF gaps of at most $G_i$ are merged; subsequently, complete ON runs shorter than $D_i$ are removed. Runs touching a sequence boundary are retained because their full duration is unobserved. Thus, $\tau_i^P$, $G_i$, and $D_i$ denote the ON-power threshold, maximum OFF-gap duration, and minimum ON duration for appliance $i$, respectively. Table I specifies their physical-time values, which are converted to 8 s samples by half-up rounding before labeling. UK-DALE house 5 uses a 30 s fridge gap.

### C. Experimental Setup

All experiments were conducted on a single NVIDIA GeForce RTX 4090 graphics processing unit (GPU; 24 GB) using PyTorch 2.6.0 with Compute Unified Device Architecture (CUDA) 12.4. MART-NILM is compared with UNet-NILM [9], the convolutional MATNilm variant (MATConv) [15], and the multi-appliance-task network of Sun *et al.* [12]. Each baseline retains the architecture, objective, and training hyperparameters of its source paper, while all models use the common house-held-out data protocol. Table II lists their settings.

TABLE I
ON–OFF LABEL PARAMETERS. THE FINAL COLUMN LISTS UK-DALE / REFIT VALUES.

| Appliance | $\tau^P$ (W) | $G$ (s) | $D$ (s) |
|---|---|---|---|
| Kettle | 200 | 0 | 12 / 12 |
| Fridge† | 50 | 12 | 60 / 60 |
| Dishwasher† | 50 | 1800 | 960 / 960 |
| Washer† | 20 | 960 | 960 / 960 |
| Microwave | 200 | 30 | 12 / 6 |

† For each sample, a five-sample local-median filter sets its label-generation power to zero when at least 60% of its four surrounding values are below 50 W and the sample exceeds both three times their median and 100 W. Endpoint windows use edge replication.

TABLE II
TRAINING WINDOWING AND BATCH SETTINGS. ALL WINDOW LENGTHS AND STRIDES ARE IN SAMPLES.

| Model | Input | Output | Stride | Batch | Optimizer | Epochs |
|---|---|---|---|---|---|---|
| UNet-NILM [9] | 100 | 1 | 1 | 128 | Adam | 50 |
| MAT-Conv [15] | 864 | 64 | 64 | 64 | Adam | 200 |
| Sun *et al.* [12] | 480 | 480 | 240 | 32 | Adam | 100 |
| MART-NILM | 1024 | 1024 | 512 | 64 | Adam | 150 |

MART-NILM uses weight decay $10^{-4}$ and unit-norm gradient clipping. Its learning rate is reduced by a factor of 0.5 after 12 epochs without validation improvement, to a minimum of $10^{-5}$. Training is limited to 150 epochs; early stopping is disabled before epoch 100 and thereafter uses a patience of 55 epochs. Its 1024-sample input and output windows each cover about 136.5 min.

For MART-NILM, checkpoint selection uses only the validation houses. Normalized mean absolute error (MAE) and average precision (AP) are computed from the validation outputs, and the retained checkpoint minimizes

$$S_{\mathrm{val}} = \mathrm{MAE}_{\mathrm{val}}^{\mathrm{norm}} + 1 - \mathrm{AP}_{\mathrm{val}}^{\mathrm{macro}}, \tag{20}$$

where $\mathrm{MAE}_{\mathrm{val}}^{\mathrm{norm}}$ is the macro mean absolute error over appliances in standardized power coordinates, and the AP term summarizes the precision–recall curve without selecting a decision threshold. For each model and appliance $i$, let $\hat{s}_{t,i} \in [0,1]$ denote the predicted ON probability at sample $t$. After checkpoint selection, an appliance-specific threshold $\gamma_i \in \{0.05, 0.06, \ldots, 0.98\}$ is selected on the validation houses to maximize appliance-level F1 after fixed temporal post-processing is applied. At test time, the decoded state label is

$$\hat{z}_{t,i} = \mathbb{I}[\hat{s}_{t,i} \geq \gamma_i], \tag{21}$$

where $\mathbb{I}[\cdot]$ is the indicator function. For the microwave, fixed temporal post-processing removes predicted ON runs shorter than 32 s and merges OFF gaps of at most 24 s. The decoded state is also applied to the power output, $\hat{P}_{t,i}^{\mathrm{eval}} = \hat{z}_{t,i}\hat{P}_{t,i}$. Thresholds and temporal rules are frozen before testing and applied unchanged to both test houses; test labels are not used to select $\gamma_i$.

### D. Evaluation Metrics

Each appliance is evaluated on power and on the on–off state. For an evaluated record of length $T_{\mathrm{rec}}$, $P_t$ is the recorded power in watts and $\hat{P}_t^{\mathrm{eval}}$ is the state-decoded prediction. The label $z_t$ equals 1 when the appliance is recorded on and 0 when it is off. $\hat{z}_t$ is the predicted label.

Results are reported separately for the held-out UK-DALE house 2 and REFIT house 20. The disjoint household protocol evaluates cross-house generalization without using any training or validation house for testing. Within each test house, the Avg. column is the macro average over the five appliances; architecture-ablation summaries give the two houses equal weight. All reported results are mean ± standard deviation over three random seeds, which quantifies optimization variability on the fixed cross-house split. The evaluation considers complementary power-estimation and state-detection metrics.

*1) Power:* Power error is reported using MAE, ON-state MAE (ON MAE), and blockwise signal aggregate error (SAE). MAE uses every sample; ON MAE is the same quantity evaluated only at samples with $z_t = 1$, so that reconstruction quality during appliance operation remains visible:

$$\mathrm{MAE} = \frac{1}{T_{\mathrm{rec}}} \sum_{t=1}^{T_{\mathrm{rec}}} \left| P_t - \hat{P}_t^{\mathrm{eval}} \right|. \tag{22}$$

Following MATNilm [15], SAE measures the mean absolute error of average power over non-overlapping 160 min blocks ($L = 1200$ samples):

$$\mathrm{SAE} = \frac{1}{K} \sum_{k=1}^{K} \left| \frac{1}{L} \sum_{t=(k-1)L+1}^{kL} \left( P_t - \hat{P}_t^{\mathrm{eval}} \right) \right|, \qquad K = \left\lfloor \frac{T_{\mathrm{rec}}}{L} \right\rfloor. \tag{23}$$

*2) On-Off State:* The recorded label $z_t$ follows the rules in Table I, and $\hat{z}_t$ is the predicted binary state. For each appliance, TP (true positives) counts samples correctly identified as on ($z_t = 1, \hat{z}_t = 1$), FP (false positives) counts inactive samples falsely detected as on ($z_t = 0, \hat{z}_t = 1$), and FN (false negatives) counts active samples missed as off ($z_t = 1, \hat{z}_t = 0$):

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{24}$$

$$\mathrm{F1} = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{25}$$

Precision is the fraction of predicted on samples that are truly on, and recall is the fraction of actual on samples detected. F1 is their harmonic mean; true negatives (TN, $z_t = 0, \hat{z}_t = 0$) are excluded to prevent long idle background periods from artificially inflating the score. To evaluate state ranking independently of the calibrated decision threshold, we also report AP from the continuous state probability. Let $\mathcal{P}_n$ and $\mathcal{R}_n$ denote precision and recall after the $n$th distinct probability threshold, with $\mathcal{R}_0 = 0$. The non-interpolated AP is

$$\mathrm{AP} = \sum_n (\mathcal{R}_n - \mathcal{R}_{n-1})\, \mathcal{P}_n. \tag{26}$$

Unlike F1, AP does not use the appliance-specific threshold or the temporal state postprocessing fitted on the validation houses.

## V. Results and Discussion

### A. Joint Power and State Estimation

Table III reports three-seed test performance; lower power errors and higher state scores are better. A displayed zero standard deviation indicates a value below the shown precision.

Across the appliance averages, MART-NILM outperforms MAT-Conv, the strongest baseline. On UK-DALE and REFIT, respectively, it reduces MAE by $23.6\%$ and $14.5\%$, ON MAE by $58.4\%$ and $34.4\%$, and SAE by $27.0\%$ and $27.8\%$; macro-F1 rises from $0.72$ to $0.88$ and from $0.58$ to $0.72$. Threshold-independent macro-AP likewise increases from $0.812$ to $0.926$ and from $0.576$ to $0.761$, with the highest displayed appliance-level AP throughout, including a tie for the UK-DALE microwave. The state gain therefore cannot be attributed solely to threshold calibration.

The advantage is not uniform across appliances. MAT-Conv remains better for UK-DALE fridge power and for REFIT kettle and microwave MAE, and its REFIT fridge F1 is marginally higher. MART-NILM instead obtains its clearest gains on dishwasher and washer estimation and on the macro averages. These exceptions show that the proposed representation improves the overall power–state balance rather than dominating every appliance–metric pair.

Figure 5 provides a qualitative view of the same comparison. MART-NILM generally follows activation boundaries and multi-stage plateaus while suppressing inactive-period power, although amplitude and timing errors remain.

### B. Performance and Model Efficiency

MART-NILM contains $1.373$ million parameters, reductions of $54.6\%$, $76.5\%$, and $82.5\%$ relative to MAT-Conv, UNet-NILM, and the network of Sun *et al.*, respectively. Together with Table III, this shows that the accuracy gains are achieved without increasing model capacity and are consistent with more structured use of shared, appliance-specific, and relational features. MART-NILM also has the smallest checkpoint and the shortest observed training and end-to-end inference times. Processing both test records requires $1.14$ s, compared with $14.67$ s for MAT-Conv, $23.50$ s for Sun *et al.*, and $80.40$ s for UNet-NILM. The benchmark includes data transfer and output reconstruction under each model's native windowing protocol, so it measures the complete prediction pipeline rather than isolated forward-pass latency. MART-NILM benefits from producing long output sequences: its 1024-sample windows advance by 512 samples, requiring substantially fewer windows than sequence-to-point inference while preserving dense temporal predictions.

### C. Architectural Contributions

Table V removes one top-level component at a time while preserving the joint five-appliance output structure and compatible feature dimensions.

The components make complementary contributions rather than improving every metric uniformly. Temporal modeling has the largest effect: removing it increases MAE by $1.44$ W and lowers macro-F1 from $0.796$ to $0.710$. Without appliance-specific decoders, ON MAE increases by $22.90$ W and macro-F1 falls to $0.783$, supporting explicit separation of appliance representations. The feature encoder gives smaller improvements across all four measures, while the multi-scale stem mainly benefits sparse, short-duration loads.

Relation attention determines the final power–state compromise. Its removal slightly improves mean ON MAE and SAE, but lowers macro-F1 from $0.796$ to $0.790$. It also increases the across-seed standard deviation from $5.35$ to $14.69$ W for ON MAE and from $0.008$ to $0.015$ for macro-F1. Thus, cross-appliance exchange does not minimize every power error; it is retained because it improves joint detection and makes the activity-sensitive results more consistent across training runs. The complete model consequently provides the best MAE and macro-F1 rather than the best value for every individual metric.

### D. Effects of the Training Strategies

Table VI evaluates focal-event mixing and appliance-wise balancing independently and jointly with the architecture fixed.

The two strategies produce distinct effects. Focal-event mixing improves macro-F1 by $0.044$ and lowers ON MAE by $56.19$ W, but increases overall MAE by $0.52$ W; it is therefore an event-coverage intervention rather than a generic regularizer. Appliance-wise balancing improves all four measures when used alone. Combining the strategies gives the best MAE, SAE, and macro-F1, raising macro-F1 by $0.082$ over standard training, whereas focal-event mixing alone retains the lowest ON MAE.

Figure 6 examines performance under different non-target backgrounds, defined as aggregate power after subtracting the five target-appliance powers. Recombining appliance events with varied residual loads is intended to reduce reliance on house-specific background correlations. Power reconstruction improves most clearly for the short-duration kettle and microwave, consistent with greater exposure to otherwise sparse events; the strong washer recall gains show that the benefit also extends to complex long-cycle loads. Some low-background combinations deteriorate, however, so the evidence supports broader background coverage rather than uniform gains.

Figure 7 addresses the complementary problem of task-scale mismatch. Under a fixed weight, the median effective state term is only $2.0\%$ and $0.8\%$ of the corresponding power loss for kettle and microwave; balancing raises these ratios to $28.5\%$ and $15.1\%$ and brings the other appliances closer to the power-loss scale. This numerical alignment is consistent with the broader validation gains in Table VI, although it represents loss-scale rather than gradient-norm balancing.

Microwave F1 changes from $0.352 \pm 0.026$ to $0.348 \pm 0.027$; no microwave detection gain is therefore claimed.

### E. Limitations and Future Work

Two limitations remain. First, microwave detection is a recurrent NILM difficulty rather than one specific to MART-NILM. Earlier neural and multi-appliance studies report weaker or highly variable microwave performance [5], [9], [15], while SAMNet shows that similar-power background

TABLE III

POWER AND ON–OFF RESULTS (MEAN ± STANDARD DEVIATION OVER THREE SEEDS). BOLD DENOTES THE BEST DISPLAYED MEAN, INCLUDING TIES.

| Metric | Method | UK-DALE | | | | | | REFIT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Kettle | Fridge | Dishwasher | Washer | Microwave | Avg. | Kettle | Fridge | Dishwasher | Washer | Microwave | Avg. |
| MAE | UNet-NILM | 9.2±2.1 | 29.1±0.9 | 22.6±5.1 | 9.1±0.1 | 6.1±0.3 | 15.2±1.1 | 8.6±0.9 | 35.0±1.7 | 12.1±1.1 | 7.6±0.1 | 7.4±0.3 | 14.2±0.3 |
| | MAT-Conv | 5.5±1.6 | **18.4±0.7** | 8.9±2.1 | 9.6±1.4 | 5.3±0.4 | 9.5±1.1 | **5.4±0.4** | 40.8±0.1 | 5.4±0.8 | 6.2±2.3 | **6.0±0.7** | 12.8±0.6 |
| | Sun *et al.* | 8.8±1.7 | 34.4±3.0 | 14.7±6.2 | 15.5±5.0 | 6.3±0.1 | 15.9±1.3 | 10.9±1.6 | 35.6±1.8 | 9.0±1.2 | 18.0±7.3 | 6.7±0.1 | 16.1±0.8 |
| | **MART-NILM** | **3.8±0.2** | 19.8±0.7 | **4.2±0.4** | **5.3±0.3** | **3.4±0.1** | **7.3±0.2** | 5.9±0.1 | **33.0±1.2** | **4.9±0.2** | **4.1±0.8** | 6.8±0.1 | **10.9±0.2** |
| SAE | UNet-NILM | 8.4±2.6 | 24.5±2.5 | 21.1±6.5 | 9.0±0.1 | 6.0±0.4 | 13.8±1.3 | 5.6±1.1 | 25.9±2.6 | 9.8±0.3 | 7.1±0.3 | 7.0±0.2 | 11.1±0.7 |
| | MAT-Conv | 4.3±1.2 | **10.0±1.0** | 6.7±1.4 | 8.8±1.3 | 5.0±0.4 | 6.9±0.8 | 2.9±0.3 | 23.7±0.6 | 4.5±1.0 | 5.0±1.8 | **5.2±0.5** | 8.3±0.5 |
| | Sun *et al.* | 7.3±1.4 | 32.1±3.5 | 12.0±5.9 | 14.5±4.9 | 6.2±0.1 | 14.4±1.4 | 7.1±1.1 | 21.7±4.4 | 7.9±1.3 | 13.0±5.5 | 6.6±0.0 | 11.3±0.7 |
| | **MART-NILM** | **2.9±0.1** | 12.8±1.5 | **2.6±0.7** | **4.0±0.4** | **2.9±0.1** | **5.1±0.4** | **2.6±0.1** | **15.4±1.6** | **3.7±0.2** | **3.1±0.9** | 5.2±0.1 | **6.0±0.2** |
| ON MAE | UNet-NILM | 1296.5±443.8 | 48.4±4.0 | 747.7±180.9 | 462.6±7.3 | 1178.8±93.3 | 746.8±110.0 | 1289.1±190.7 | 63.4±3.4 | 398.3±93.1 | 306.1±6.7 | 1196.7±58.0 | 650.7±59.1 |
| | MAT-Conv | 547.5±180.2 | **22.1±1.6** | 206.9±46.2 | 385.7±59.1 | 946.4±99.8 | 421.7±75.2 | 850.7±44.4 | 46.3±2.5 | 208.2±21.0 | 195.6±10.1 | 1062.2±90.6 | 472.6±29.3 |
| | Sun *et al.* | 807.3±69.1 | 61.0±7.1 | 449.8±212.8 | 478.2±28.0 | 1215.2±19.2 | 602.3±44.3 | 1012.0±65.7 | 59.1±6.4 | 437.1±91.3 | 306.7±17.6 | 1224.2±5.6 | 607.8±16.5 |
| | **MART-NILM** | **383.4±15.9** | 25.4±2.1 | **87.0±7.4** | **161.8±17.0** | **218.8±23.9** | **175.3±6.9** | **785.2±17.4** | **40.4±1.5** | **139.5±6.0** | **155.9±5.3** | **429.0±40.8** | **310.0±3.5** |
| AP | UNet-NILM | 0.908±0.016 | 0.873±0.041 | 0.580±0.274 | 0.866±0.026 | 0.535±0.024 | 0.753±0.049 | 0.653±0.036 | 0.657±0.030 | 0.269±0.056 | 0.654±0.129 | 0.136±0.046 | 0.474±0.024 |
| | MAT-Conv | 0.968±0.020 | 0.951±0.006 | 0.868±0.039 | 0.555±0.191 | **0.718±0.063** | 0.812±0.051 | 0.801±0.018 | 0.537±0.005 | 0.736±0.040 | 0.653±0.066 | 0.152±0.046 | 0.576±0.024 |
| | Sun *et al.* | 0.930±0.016 | 0.915±0.030 | 0.838±0.094 | 0.207±0.128 | 0.514±0.194 | 0.681±0.075 | 0.660±0.090 | 0.641±0.053 | 0.615±0.040 | 0.123±0.019 | 0.160±0.069 | 0.440±0.000 |
| | **MART-NILM** | **0.997±0.000** | **0.970±0.001** | **0.959±0.002** | **0.985±0.007** | **0.722±0.046** | **0.926±0.010** | **0.848±0.007** | **0.736±0.030** | **0.759±0.014** | **0.914±0.010** | **0.547±0.056** | **0.761±0.016** |
| Precision | UNet-NILM | 0.80±0.09 | 0.74±0.09 | 0.55±0.19 | 0.84±0.09 | 0.57±0.07 | 0.70±0.03 | 0.63±0.08 | 0.60±0.12 | 0.26±0.06 | **0.89±0.05** | 0.18±0.07 | 0.51±0.04 |
| | MAT-Conv | 0.82±0.10 | 0.76±0.03 | 0.47±0.05 | 0.61±0.18 | **0.71±0.09** | 0.67±0.08 | 0.70±0.10 | 0.59±0.00 | 0.42±0.06 | 0.59±0.07 | 0.16±0.03 | 0.49±0.04 |
| | Sun *et al.* | 0.74±0.08 | 0.87±0.01 | 0.51±0.16 | 0.22±0.09 | 0.55±0.49 | 0.58±0.11 | 0.57±0.08 | 0.63±0.00 | 0.54±0.09 | 0.14±0.02 | 0.37±0.28 | 0.45±0.04 |
| | **MART-NILM** | **0.93±0.01** | **0.89±0.02** | **0.65±0.03** | **0.92±0.01** | **0.72±0.00** | **0.82±0.00** | **0.74±0.01** | **0.64±0.01** | **0.63±0.02** | 0.83±0.07 | **0.46±0.02** | **0.66±0.01** |
| Recall | UNet-NILM | 0.87±0.08 | 0.82±0.14 | 0.65±0.21 | 0.78±0.04 | 0.57±0.07 | 0.74±0.05 | 0.73±0.03 | 0.65±0.18 | 0.46±0.07 | 0.43±0.13 | 0.37±0.04 | 0.53±0.07 |
| | MAT-Conv | 0.97±0.01 | **0.98±0.02** | 0.98±0.02 | 0.49±0.18 | 0.67±0.13 | 0.82±0.02 | 0.78±0.03 | **0.93±0.06** | **0.83±0.02** | 0.76±0.21 | 0.42±0.25 | 0.74±0.11 |
| | Sun *et al.* | 0.92±0.03 | 0.94±0.01 | 0.89±0.14 | 0.56±0.06 | 0.08±0.07 | 0.68±0.05 | 0.75±0.05 | 0.78±0.01 | 0.77±0.08 | 0.50±0.12 | 0.03±0.02 | 0.57±0.01 |
| | **MART-NILM** | **1.00±0.00** | 0.94±0.01 | **0.99±0.01** | **0.97±0.01** | **0.90±0.02** | **0.96±0.00** | **0.82±0.01** | 0.78±0.04 | 0.75±0.03 | **0.88±0.03** | **0.72±0.04** | **0.79±0.02** |
| F1 | UNet-NILM | 0.83±0.03 | 0.77±0.07 | 0.60±0.20 | 0.81±0.03 | 0.57±0.04 | 0.71±0.03 | 0.67±0.03 | 0.61±0.08 | 0.33±0.07 | 0.58±0.12 | 0.23±0.05 | 0.48±0.04 |
| | MAT-Conv | 0.89±0.06 | 0.86±0.01 | 0.63±0.04 | 0.54±0.17 | 0.68±0.06 | 0.72±0.05 | 0.73±0.05 | **0.72±0.02** | 0.55±0.05 | 0.65±0.06 | 0.23±0.08 | 0.58±0.02 |
| | Sun *et al.* | 0.82±0.04 | **0.91±0.00** | 0.64±0.14 | 0.30±0.09 | 0.15±0.13 | 0.56±0.05 | 0.64±0.03 | 0.70±0.00 | 0.63±0.05 | 0.21±0.01 | 0.06±0.04 | 0.45±0.00 |
| | **MART-NILM** | **0.96±0.00** | **0.91±0.00** | **0.79±0.02** | **0.94±0.01** | **0.80±0.01** | **0.88±0.00** | **0.78±0.01** | 0.71±0.01 | **0.69±0.00** | **0.85±0.03** | **0.56±0.03** | **0.72±0.00** |

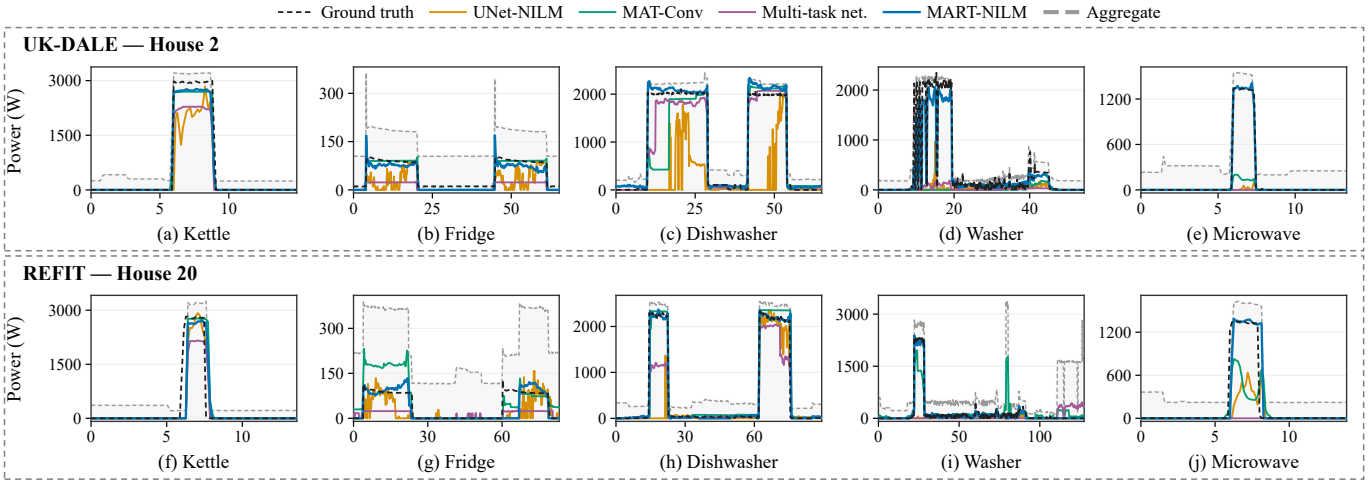



Fig. 5. Waveform comparisons on the two test houses.

TABLE IV

MODEL SIZE AND OBSERVED COMPUTATIONAL COST. INFERENCE TIME COVERS BOTH TEST RECORDS AT BATCH SIZE 64 UNDER EACH MODEL'S NATIVE WINDOWING PROTOCOL. MB DENOTES MEGABYTES.

| Model | Parameters (M) | Model size (MB) | Training (s) | Inference (s) |
|---|---|---|---|---|
| UNet-NILM | 5.832 | 22.299 | 17455.8 | 80.40 |
| MAT-Conv | 3.022 | 11.611 | 10354.9 | 14.67 |
| Sun *et al.* | 7.829 | 59.793 | 16493.8 | 23.50 |
| MART-NILM | **1.373** | **5.335** | **485.2** | **1.14** |

TABLE V

ARCHITECTURE ABLATION (THREE-SEED MEAN ± STANDARD DEVIATION OVER THE TWO TEST HOUSES).

| Variant | Parameters (M) | MAE (W) ↓ | ON MAE (W) ↓ | SAE (W) ↓ | Macro-F1 ↑ |
|---|---|---|---|---|---|
| Single-scale CNN | 1.372 | 9.22 ±0.18 | 241.75 ±8.76 | 5.66 ±0.17 | 0.795 ±0.001 |
| Without feature encoder | 1.326 | 9.16 ±0.06 | 252.01 ±2.21 | 5.59 ±0.07 | 0.790 ±0.006 |
| Without temporal encoder | 0.634 | 10.56 ±0.33 | 288.07 ±10.25 | 6.56 ±0.16 | 0.710 ±0.013 |
| Without appliance decoders | 0.836 | 9.22 ±0.66 | 265.84 ±4.30 | 5.63 ±0.63 | 0.783 ±0.015 |
| Without relation attention | 1.332 | 9.18 ±0.09 | **230.84 ±14.69** | **5.55 ±0.04** | 0.790 ±0.015 |
| **MART-NILM** | 1.373 | **9.12 ±0.14** | 242.94 ±5.35 | 5.58 ±0.17 | **0.796 ±0.008** |

loads can be mistaken for the target appliance [14]. Its short, intermittent, and variable-power activations compound this ambiguity. Focal-event mixing mitigates the problem across several residual-power conditions in Fig. 6, but does not remove it. Second, performance remains sensitive to the residual-background level: under high aggregate demand, overlapping non-target activity can obscure appliance transitions even when the target signature is unchanged. Domain variation

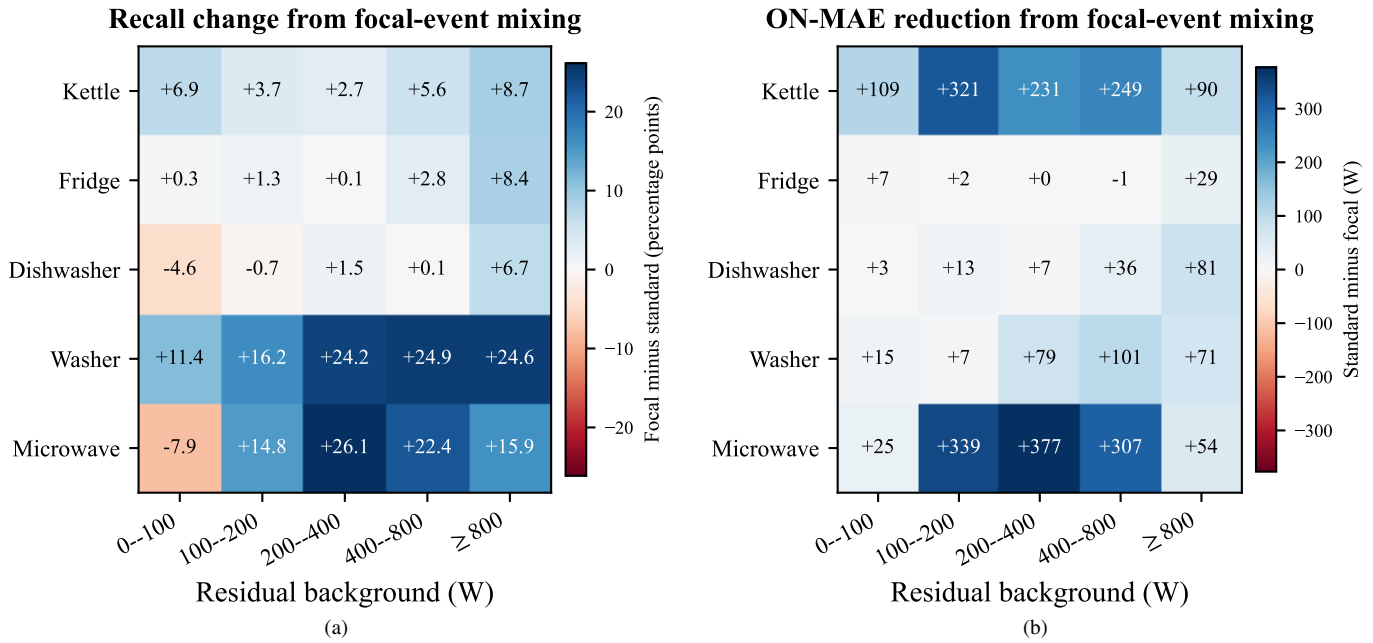



Fig. 6. Effect of focal-event mixing across validation residual-power intervals: (a) active-state recall change in percentage points; (b) ON MAE reduction in watts. Positive values indicate improvement.

TABLE VI
TRAINING-STRATEGY ABLATION ON THE VALIDATION HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION). † DENOTES A STRATEGY PROPOSED IN THIS WORK.

| Training strategy | MAE (W) ↓ | ON MAE (W) ↓ | SAE (W) ↓ | Macro-F1 ↑ |
|---|---|---|---|---|
| Standard joint training | 21.164 ±0.394 | 357.808 ±19.280 | 15.938 ±0.348 | 0.605 ±0.012 |
| + Focal-event mixing† | 21.679 ±0.144 | **301.623 ±16.196** | 15.601 ±0.365 | 0.649 ±0.016 |
| + Appliance-wise balancing† | 20.929 ±0.065 | 334.862 ±10.888 | 15.175 ±0.322 | 0.661 ±0.014 |
| **Complete MART-NILM†** | **20.791 ±0.175** | 315.408 ±10.405 | **15.016 ±0.385** | **0.687 ±0.003** |

in NILM should therefore encompass both appliance-signature shifts and changes in household background composition. Future work should model these two sources jointly across houses and regions rather than treating appliance waveform variation alone.

## VI. CONCLUSION

This paper proposed MART-NILM and two complementary training strategies for joint appliance power and state estimation from low-frequency aggregate measurements. The architecture extracts shared temporal evidence, separates appliance-specific representations, and exchanges relational context before state-gated power prediction. To support learning from sparse activations, focal-event background-stratified mixing varies the residual demand surrounding appliance events, while bounded appliance-wise balancing coordinates their power and state objectives. Both strategies operate during training without additional inference computation. Across three seeds, MART-NILM achieves appliance-average MAEs of 7.3 and 10.9 W and macro-F1 scores of 0.88 and 0.72 on the UK-DALE and REFIT test houses, respectively, outperforming

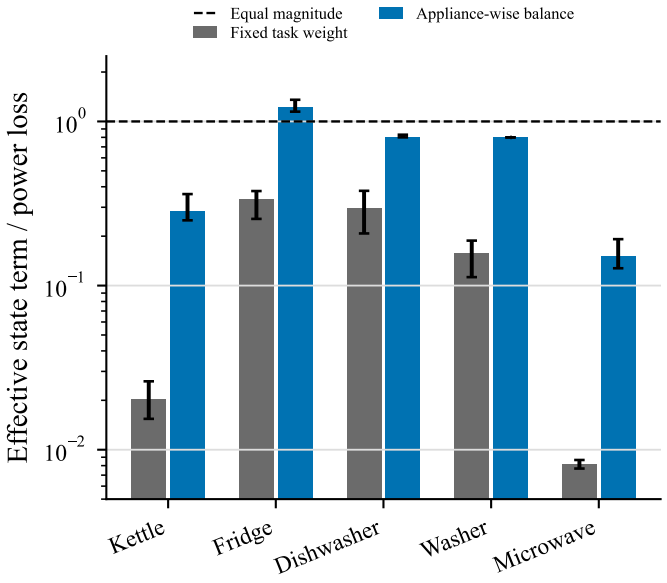



Fig. 7. Power–state scale alignment produced by appliance-wise balancing. Bars show the median effective state-term–to–power-loss ratio pooled over epochs 20 onward and three seeds; error bars denote the interquartile range.

three reproduced baselines on both measures with 1.373 million parameters. The ablations support the complementary roles of temporal modeling, appliance specialization, and the training strategies. Together, these findings support compact joint power–state estimation as a smart-meter observation layer for residential demand-side monitoring.

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